

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

An Online Digital Ordering System With Membership Management

A Capstone Project Proposal

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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Second Semester
Academic Year 2025–2026

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The rapid advancement of digital technology in a recent years has fundamentally altered how businesses operate and interact with their customers. The expansion of the internet and mobile applications has made it possible for business transition from manual to automated digital systems. One of the biggest developments in the use of online ordering platforms, which let customers browse products and services, place orders, and complete transactions without needing to interact in person.

The advent of the internet and emerging technologies has significantly transformed business operations, particularly for medium-sized enterprises in the Philippines. These businesses must enhance operational efficiency, customer satisfaction, and overall performance, with robust order management and inventory tracking systems playing a pivotal role.

Amazing World Marketing Corporation (AWMC), a direct-selling company, offers a diverse product lineup including wellness items, personal care products, health supplements such as Immukira-AG and Immuvit-ZinC, and Ardeur de France fragrances. It features tiered membership levels from Silver to Diamond, providing exclusive benefits and business packages to members. Currently, manual order processing leads to errors, inventory discrepancies, and delays.

Traditional ordering systems, which are frequently used in small and medium-sized enterprises, frequently depend on manual procedures like verbal interaction, handwritten orders, and actual payment methods. These methods are prone to delays, inefficiencies, and human mistake, which could result in disappointed consumers. Also, business owners find it challenging to measure sales, keep an eye on consumer behavior, and provide accurate reports when using manual methods due to improper data management.

Businesses are increasingly putting online digital ordering systems and into place to deal with the issues. By automating order processing, lowering errors, and increasing transaction speed, this systems optimize operations. While businesses gain greater control over their operations and data, customers benefit from the ease of ordering an anytime and from any location. Membership management has become a crucial part of contemporary business strategies alongside the development of ordering systems. Businesses can register users, store client data, and provide individualized services with membership systems. Reward points, discounts, special offers, and loyalty programs are examples of features that promote repeat business and improve customer relations. Combining membership management with an online ordering system offer a complete solution that improve customer engagement and operational effectiveness. Businesses can provide a smooth user experience through this interaction while gathering useful data for make marketing and decision making.

The proposed web-based AWMC Ordering System, developed using PHP, MySQL, HTML, CSS, and JavaScript, addresses these issues by enabling diverse payment methods like GCash, PayMaya, and PayPal. It includes dedicated interfaces for administrators and customers, automated queue numbering, and an analytics dashboard for data-driven decisions. This project applies academic knowledge in web development, database management, and systems analysis, exemplifying digital transformation for AWMC.

1.2 Statement of the Problem

AWMC is currently facing several operational challenges as a result of the lack of comprehensive and integrated online organ system. The reliance on manual processes and fragmented tools has led to inefficiencies that have an impact on both business operations and customer satisfaction. A businesses increasingly turn two digital solutions to stay competitive AWMC's lack of such a system puts it an disadvantage in terms of service quality, operational control, and customer engagement.

The primary issues include:

- Lack of a centralized platform for managing and tracking customer orders. Currently, orders are handled manually or through unstructured communication channels, which increases the possibility of human error. These errors can include incorrect order information, missed transactions, or order duplication. Furthermore, without a proper tracking system, it is difficult to monitor the status of each order,

resulting in delays in processing and delivery. This lack of transparency has a negative impact on both employees and customers, causing confusion and inefficiency in day-to-day operations.

- Inventory management is critical to ensuring that products are available when customers place orders. However, without an automated system, stock levels are frequently updated manually, which leads to inaccuracies. This can result in situations where items appear to be available but are actually out of stock, and vice versa. Furthermore, the lack of real-time inventory monitoring makes it difficult for management to identify low-stock items in a timely manner, potentially resulting in missed sales opportunities and customer dissatisfaction.

- Membership programs are essential in today's business environments because they foster customer loyalty and encourage repeat transactions. Without a structured membership system, AWMC cannot efficiently manage customer data, track member activity, or offer personalized benefits like discounts, rewards, or exclusive promotions. This limitation impairs the company's ability to retain customers and build long-term relationships with its clients.

- AWMC provides limited payment options, which limits customer convenience and flexibility. Customers in today's digital age expect a variety of payment options, such as digital wallets and online payment platforms. The lack of such options may deter potential customers from completing their transactions, lowering overall sales performance. Diverse payment methods are essential for catering to customers' varying preferences and improving their overall experience.

- Lack of real-time sales tracking and reporting. Without an automated system, sales data is frequently recorded manually and analyzed on a regular basis, which slows down decision-making processes. Managers do not have access to current information on revenue, order trends, and product performance.

- This limitation impedes their ability to make informed decisions about inventory replenishment, pricing strategies, and business planning. Real-time analytics are critical for identifying trends, forecasting demand, and increasing operational efficiency.

- AWMC experiences inefficiencies in order queuing, especially during peak hours. The absence of a structured queuing system leads to disorganized order processing, which causes confusion among employees and customers. Customers may face lengthy wait times, unclear order status, and dissatisfaction during pickup or delivery.

An inefficient queuing process not only degrades the customer experience, but it also reduces overall business productivity.

- A system like this should streamline order processing, automate inventory management, support membership features, provide multiple payment options, enable real-time analytics, and implement an efficient waiting mechanism. Resolving these issues will allow AWMC to significantly improve its operational efficiency, increase customer satisfaction, and strengthen its market competitiveness.

1.3.2 Specific Objectives

General Objective

To develop a web-based ordering system for AWMC that automates order processing, inventory management, membership handling, payments, queuing, and analytics. The proposed system aims to enhance operational efficiency, improve accuracy, and provide a better overall experience for both customers and administrators.

Specific Objectives

The study specifically aims to:

- The initial goal is to identify the proper system specifications and technical needs by analyzing AWMC's current operations and user requirements. This involves looking at the procedures currently in place for managing inventory, processing orders, and communicating with clients. The study intends to create a solution that directly deals with the errors and inefficiencies in the current system. In order to make sure that the system satisfies their requirements and expectations, this phase also entails getting feedback from stakeholders like company owners, staff members, and clients.
- Creating a responsive and user-friendly online interface that serves administrators and customers is the second goal. To guarantee that every kind of user gets access to the right features, the system will have unique dashboards. While administrators will have capabilities to manage orders, inventory, and system settings, customers will be able to browse products, place purchases, and maintain their accounts. To guarantee accessibility, usability, and responsiveness across many devices, the interface will be created utilizing contemporary web technologies including HTML, CSS, and JavaScript.
- The final goal is to use MySQL and PHP to create a reliable backend system. Core functions like order processing, inventory tracking, user authentication, and

membership management will be handled by this backend. Data will be managed and stored securely by the system, guaranteeing accuracy, consistency, and ease of retrieval. Additionally, the backend will be built with scalability in mind, enabling the system to handle future expansion and new features.

- Another goal is to use an automated queue numbering system to improve order processing efficiency. This feature assigns queue numbers to customer orders, allowing both employees and customers to monitor the status of each transaction. Implementing a structured queuing system will help to reduce confusion, shorten wait times, and improve the overall customer experience for pickup and delivery operations.
- The study also aims to incorporate multiple payment gateways to provide customers with a variety of payment options. These include digital payment platforms such as GCash, PayMaya, and PayPal, as well as traditional methods like cash on pickup and cash on delivery. By offering flexible payment options, the system will cater to a wider range of customers and improve transaction completion rates.
- The study also aims to create an administrative dashboard with real-time sales data. Key performance metrics like total revenue, order volume, best-selling items, and membership conversion rates will be shown on this dashboard. Managers will be able to effectively monitor overall performance, optimize business strategies, and make well-informed decisions thanks to the availability of real-time data.
- In order to make sure that the developed system satisfies quality standards, the study's final goal is to carry out thorough system testing. Functionality, usability, security, performance, and dependability testing are all included in this. Usability testing will make sure the system is user-friendly, while functional testing will confirm that all features operate as intended. Performance testing will assess the system's capacity to manage numerous users and transactions, while security testing will shield private information from unwanted access. Testing for reliability will guarantee that the system performs consistently in various scenarios.

1.4 Scope and Delimitation

The system focuses on AWMC's core operations: order placement, inventory tracking, membership management, payments, queuing, and analytics. It utilizes specific web technologies such as HTML, CSS, JavaScript, PHP, and MySQL to ensure functionality, responsiveness, and efficient data handling. It targets AWMC

staff and customers in the Philippines, using specified technologies. Delimitations include no integration with external logistics or advanced AI features;

1.5 Significance of the Study

This system benefits multiple stakeholders.

- AWMC Management and Staff:** Order processing automation minimizes human error and increases accuracy by reducing the need for manual input. The system's ability to track stock levels in real time makes inventory management more effective by assisting employees in preventing shortages or overstocking. The system also produces comprehensive sales reports and analytics, which enable management to track company performance and spot trends. These insights facilitate well-informed decision-making, allowing the company to create successful growth and sustainability strategies. Additionally, employees can concentrate on higher-value tasks due to the decrease in manual labor, which boosts overall productivity and service quality.

- Customers:** Customers don't need to visit a physical location to browse available products, place orders, and choose from a variety of payment options thanks to the web-based platform. Digital payment options like PayPal, PayMaya, and GCash are included to provide flexibility and accommodate a range of consumer preferences. Customers can also track the status of their purchases in real time with the system's order tracking features. By providing exclusive access to specific products, discounts, and rewards, membership functionality adds even more value, boosting customer loyalty and promoting repeat business. All things considered, the system makes shopping more effective, convenient, and fulfilling.

- System Administrators:** Provides a centralized, organized platform for managing corporate operation. Through an easy-to-use dashboard, administrators can manage customer accounts, keep an eye on inventory, oversee orders, and analyze performance metrics. Administrators can maintain seamless system performance and react swiftly to operational problems with real-time data visibility. Additionally, the system lowers the risks connected with manual record-keeping by ensuring improved data organization and security. The system helps to improve operational control and system management by streamlining administrative tasks.

- Future Researchers and Students:** This study is a useful academic and practical resource. It illustrates how database systems and web development technologies can be used to solve practical business issues. Students studying information technology in particular can gain from knowing how various elements-like database management, backend logic, and frontend interfaces combine to form a functional

system. Additionally, the study offers insights into system design, implementation, and evaluation that can serve as a basis for further investigations. Additionally, it promotes innovation by demonstrating how digital solutions can improve customer engagement and business processes.

1.6 Definition of Terms

•AWMC Ordering System: The AWMC Ordering System refers to a web-based ordering and management platform developed in this study to support the business operations of AWMC. The system enables customers to browse available products, place online orders, select payment methods, and track their transactions efficiently. It is designed to provide convenience and accessibility for customers who prefer digital purchasing methods. The platform also contains queue management features that help organize customer orders according to their sequence and priority. In addition, the system includes sales analytics tools that allow administrators to monitor revenue, transactions, and product performance. Administrative controls are integrated into the system to manage users, products, orders, and reports effectively. The ordering system minimizes manual processes and improves operational efficiency within the business. It also provides a user-friendly environment where customers can interact with the platform smoothly and securely. The system supports both customer-side and administrator-side operations through interconnected modules. Overall, the AWMC Ordering System serves as the primary technological solution developed to modernize the ordering and transaction processes of AWMC.

• Member: A registered user who has successfully created an account in the AWMC Ordering System and is active in the membership program is referred to as a member. Members can access to exclusive features and benefits that are not available to visitors. Members are divided into Silver, Gold, Platinum, and Diamond categories by the system based on their membership requirements or buying activity. Rewards, discounts, and promotional advantages are offered in proportion to each membership level. Additionally, members could get special announcements from the firm and access to unique products. Customer loyalty and ongoing transactions on the site are encouraged by the membership function. Through their accounts, registered users may keep track of their order history, modify their personal profiles, and keep an eye on their transactions. To ensure accurate records, membership data is safely kept in the system database. In order to apply the appropriate rights and discounts, the system automatically detects member status during transactions. All things considered, the membership program improves client interaction and fortifies enduring ties between clients and AWMC.

- **Queue Number:** Within the AWMC Ordering System, each client order is automatically issued a Queue Number. The order processing sequence is arranged and tracked in an orderly fashion using this identity. In accordance with the sequence in which they were received, queue numbers help guarantee that client transactions are processed equally and carefully. To prevent misunderstanding and duplication during processes, the system automatically creates unique queue numbers. Consumers may monitor the progress of their orders and ascertain their position in the processing order by using their queue numbers. Queue numbers are also used by administrators and employees to effectively handle pickups, deliveries, and transaction verification. The queue management tool enhances service organization and shortens client wait times. Additionally, it reduces the possibility of mistakes in manual order handling procedures. For the sake of documentation and transaction tracking, queue numbers are kept in the database. In general, the queue number system improves customer happiness and operational flow.

- **Middleware:** The PHP-based elements of the AWMC Ordering System that serve as a bridge between the frontend interface and the backend database are referred to as middleware. It is in charge of handling requests and replies amongst various system tiers. By managing crucial tasks in the background, middleware guarantees seamless communication between users and the server. User authentication, which confirms users' identities before allowing access to system functionalities, is one of its main features. In order to ensure safe and ongoing interactions while users explore the site, middleware also controls user sessions. It also manages permission procedures to guarantee that users may only access features that are appropriate for their designated responsibilities. Another crucial duty of middleware is data validation, which verifies that user inputs are accurate, safe, and complete before storing them in the database. By limiting system vulnerabilities and blocking unwanted access, middleware enhances the system's overall security and dependability. Additionally, it increases the platform's connectivity and data processing efficiency. All things considered, middleware acts as a crucial link between the user interface and database and server functions.

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- **API:** The collection of PHP-based interfaces that allow communication between various AWMC Ordering System components is known as the API, or Application Programming Interface. The system's frontend and backend may communicate data and carry out tasks effectively thanks to APIs. Customers may place orders, add items to their carts, and get real-time transaction data via APIs. Additionally, APIs are in charge of syncing transaction data throughout the platform and updating order statuses. APIs are used by the administration dashboard to present current and accurate sales and customer data. By establishing precise guidelines and directives for data interchange, APIs streamline communication between system elements. By automating tedious tasks and minimizing human interaction, they enhance system

performance. Additionally, APIs increase the platform's scalability and adaptability to upcoming system improvements or third-party integrations. Sensitive client and transaction data is protected by security protections implemented into APIs. All things considered, APIs are crucial to the smooth operation and communication of the AWMC Ordering System.

- **Database Gateway:** The AWMC Ordering System's MySQL-based data management layer is known to as the Database Gateway. It is in charge of effectively organizing, retrieving, storing, and managing system data. Records pertaining to users, goods, transactions, orders, membership information, and payment details are all included in the database. Every time a user interacts with the platform, the database gateway allows the system to swiftly access and process data. By keeping correct and consistent data throughout system activities, the database guarantees data integrity. Additionally, it offers safe storage options to prevent unwanted access to client and company data. Administrators may methodically update, remove, and manage records thanks to the database gateway. Improved system speed and quicker transaction processing are two benefits of effective database administration. To guarantee system dependability and avoid data loss, backup and recovery procedures may also be used. In general, the database gateway functions as the AWMC Ordering System's core information management and storing component.

- **Ardeur de France:** Ardeur de France is the name of AWMC's exclusive perfume line that is included in the ordering system. The perfume items are made to appeal to both men and women and are influenced by well-known scent brands worldwide. One of the primary product lines that may be purchased using the AWMC Ordering System is this one. Within the portal, customers may peruse a variety of scent variations, descriptions, and price information. The system can digitally display AWMC's distinctive items thanks to the incorporation of Ardeur de France products. The company's whole sales and marketing strategy benefits from the perfume line. Online purchases and delivery options make it easy for customers to order Ardeur de France items. The administrative side of the system is used to handle inventory information and product availability. Additionally, the platform enables discounts and promotional offers to be applied to specific scent goods. All things considered, the main highlighted product category incorporated into the AWMC Ordering System is Ardeur de France.

- **GCash/PayMaya/PayPal:** Digital payment systems like GCash, PayMaya, and PayPal are included into the AWMC Ordering System to facilitate safe and practical online purchases. With the help of these payment options, consumers may make purchases online without using actual currency. While PayPal is well-known worldwide for online payments, GCash and PayMaya are popular electronic wallet

services in the Philippines. Customers' convenience and flexibility during transactions are improved by the integration of different payment methods. Consumers may use their internet accounts or mobile devices to safely transfer money. Digital payments increase transaction efficiency and eliminate the need for in-person cash handling. For monitoring reasons, the system automatically logs transaction data and payment confirmations. To protect financial data and stop illegal activities, security procedures are put in place. Customers have additional alternatives based on their preferences thanks to the availability of numerous payment methods. In general, the use of digital payment systems enhances the entire consumer experience and modernizes the payment process.

- **Cash on Pickup (COP):** The AWMC Ordering System offers a payment option called Cash on Pickup, or COP. When consumers pick up their orders from the specified pickup location, they can use this payment option to pay for their products in cash. Customers that prefer cash transactions over digital payment methods are the target market for COP. Customers who might not have access to online banking or e-wallet services can benefit from the method's flexibility. Customers may still take advantage of the ease of online purchasing with COP, even if they need to pay in person upon pickup. For appropriate order management, the system logs the chosen payment method and transaction data. During the pickup procedure, employees check the order and take payment. This method of payment contributes to greater consumer inclusion and accessibility. For some consumers, it also lowers the possibility of unsuccessful online payment operations. All things considered, Cash on Pickup is a useful substitute payment option in the AWMC Ordering System.

- **Cash on Delivery (COD):** Cash on Delivery, or COD, is a payment method that lets clients pay with cash when their products are delivered. Customers who would rather check their orders before paying might choose this payment option. COD offers simplicity and flexibility, particularly for consumers who don't utilize digital payment systems. Customers can use this option to conduct online orders and pay only when the products are delivered. Payment collection and transaction verification fall within the purview of delivery staff. After a successful delivery, the system changes the payment status and logs COD transactions. By allowing payment upon actual goods receipt, COD contributes to the development of client confidence. Additionally, it makes the system accessible to clients in places where internet payments are not widely used. To prevent delivery problems or transaction disputes, appropriate monitoring and verification processes are put in place. All things considered, Cash on Delivery provides a convenient payment option inside the AWMC Ordering System.

- **Sales Analytics:** The act of gathering, evaluating, and displaying sales-related data inside the AWMC Ordering System is known as sales analytics. Administrators can keep an eye on crucial company data including total income, transaction volume, and consumer purchasing patterns using this functionality. Effective management evaluation of business performance is made possible by sales analytics' visual reports and summaries. The best-selling items and consumer purchase patterns can be identified by the system. Better decision-making and strategy planning for the company are supported by these insights. Administrators can keep an eye on the success of membership programs, discounts, and promotions with the use of sales analytics. In business analysis, accurate reporting minimizes human error and minimizes the need for manual computation. By organizing sales data into understandable formats such as tables and charts, the feature increases operational efficiency. It is also important to keep historical sales records for comparison and future use. All things considered, sales analytics is a crucial management tool that helps AWMC make data-driven decisions.

- **User Interface (UI):** The User Interface, commonly referred to as UI, is the visual component of the AWMC Ordering System that allows users to interact with the platform. It includes elements of design such as dashboards, menus, buttons, forms, and navigation bars. Customers and administrators will consider the user interface to be straightforward, well-structured, and easy to use. Consumers may place orders, manage their carts, explore items, and keep track of transactions via the user interface. To manage items, keep an eye on sales, and update transaction data, administrators engage with the interface. A well-designed user interface promotes effective system utilization and increases accessibility. By making system operations simpler to comprehend and use, the user interface (UI) also improves client satisfaction. To ensure compatibility across computers, tablets, and mobile devices, responsive design techniques can be used. Organized design and visual consistency reduce user confusion during interactions. In general, users interact with the AWMC Ordering System primarily through the User Interface.

- **Authentication:** The process of verifying a user's identity before letting them to use the AWMC Ordering System is known as authentication. Typically, this procedure calls for users to supply login information, including passwords, email addresses, and usernames. Only those who are permitted can access the system and its functions thanks to authentication. To verify the authenticity of user credentials, the system compares them with database entries that have been saved. Sensitive data and client accounts are shielded from unwanted access by secure authentication processes. Session management and encrypted passwords are two further security techniques that might be used. Maintaining the system's confidentiality and integrity requires authentication. Before they may access transaction data and personalized services, registered users must successfully complete the authentication procedure.

In order to use the platform's administration and monitoring features, administrators must also go through authentication. All things considered, authentication is a vital security component that safeguards both the AWMC Ordering System and its users.

- **Authorization:** The process of determining out the degree of access and permissions given to users inside the AWMC Ordering System is referred to as authorization. After authentication is successful, the system determines the user's assigned role and restricts access appropriately. Within the platform, several user roles—such as administrators and customers—have distinct duties and rights. Customers can only place orders and access personal account information, but administrators may manage items, keep an eye on sales figures, and adjust transaction statuses. By prohibiting unauthorized users from accessing restricted functions, authorization contributes to system security. Authorization protocols are used by the system to restrict access to management tools and sensitive data. Furthermore to enhancing accountability, proper permission lowers the possibility of deliberate or inadvertent system abuse. The system's middleware and database functions incorporate authorization procedures. All things considered, authorization is a crucial tool for regulating user access and upholding safe system operations.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Literature

Advancements in electronic commerce, management information systems, and software engineering principles have all had a significant impact on the evolution of modern business systems. Foreign literature provides a solid theoretical and conceptual foundation for understanding how digital ordering systems and membership management platforms can help businesses improve their operations and customer engagement.

Electronic Commerce 2018, authored by Turban et al., posits that digital ordering systems enhance cost-efficiency and operational effectiveness. These systems

improve customer satisfaction through faster order fulfillment and reduced errors, while e-commerce extends beyond transactions to facilitate communication and information exchange—principles integral to the AWMC Ordering System.

Turban et al. emphasize that e-commerce platforms are essential for improving customer satisfaction in addition to efficiency. A better user experience is influenced by quicker order fulfillment, precise transaction processing, and enhanced communication channels. Customer engagement and trust are further enhanced by the ability to track orders and get real-time updates. The AWMC system incorporates these features to guarantee that clients receive dependable and clear service.

Management Information Systems (Laudon & Laudon)

Kenneth C. Laudon and Jane P. Laudon emphasize that robust management information systems (MIS) are essential for informed business decision-making. Such systems enable real-time analysis of sales data and reporting, directly informing the analytics dashboard in the AWMC Ordering System.

Laudon and Laudon emphasize that by centralizing data and simplifying procedures, information systems enhance organizational coordination and control. The use of a centralized system in the context of AWMC guarantees the effective recording and management of all transactions, inventory updates, and customer interactions. This improves consistency across various business operations and minimizes redundancy.

Software Engineering (Pressman & Maxim)

Roger S. Pressman and Bruce R. Maxim advocate structured methodologies for web application development, including modular decomposition, iterative refinement, and design patterns. The AWMC Ordering System incorporates these practices to ensure scalability and sustained performance.

Pressman and Maxim, is the process of developing systems in small steps with ongoing testing and improvement. This guarantees that mistakes are found and fixed early in the development process, making the system more dependable and effective. The system's overall quality is also improved by using design standards and best practices, which guarantee that it satisfies both functional and non-functional requirements.

Management Information Systems (O'Brien & Marakas)

James A. O'Brien and George M. Marakas highlight that automated inventory management systems prevent overstocking or stock outs by providing real-time tracking and updates. This functionality is central to the AWMC Ordering System, mitigating stock discrepancies and supporting efficient operations.

By highlighting the function of automated inventory management systems in contemporary businesses, O'Brien and Marakas add even more to the body of literature. They contend that maintaining operational effectiveness and customer satisfaction requires precise inventory tracking. Businesses can avoid overstocking and stockouts in the future by using automated systems that provide real-time stock level updates. In addition to cutting expenses, this guarantees that consumers can quickly obtain the goods they require.

2.2 Foreign Studies

Pavlou and Fygenson (2006)

Using Ajzen's Theory of Planned Behavior, Pavlou and Fygenson's study, which was published in MIS Quarterly, looked at user adoption of e-commerce platforms. The study found that consumer purchasing decisions are significantly influenced by trust in payment security and website dependability. Customers who believe a system is safe and reliable are more likely to finish online transactions, according to the study. Customers' opinions about online buying are greatly influenced by a website's dependability, usability, and usefulness. Customers who believe a system is safe and reliable are more likely to finish online transactions, according to the study. Customers' opinions about online buying are greatly influenced by a website's dependability, usability, and usefulness.

Pavlou and Fygenson go on to say that in order to win over customers, online payment systems need to offer security and transparency. If consumers think their personal information might be compromised, they are reluctant to divulge financial information. By including safe payment methods like online payment gateways and digital wallets, the AWMC Ordering System takes these results into account. GCash, PayMaya, PayPal, Cash on Pickup, and Cash on Delivery are just a few of the payment options available to customers.

Also, the study emphasizes how user-friendly interfaces boost consumer satisfaction and encourage repeated use. Customers are discouraged from completing transactions by difficult-to-use systems. This problem is addressed by the AWMC Ordering System, which offers accessible transaction capabilities, well-organized

menus, and easy navigation. The importance of perceived usefulness is another important finding in the study. If customers believe that online tools are more helpful and save time, they are more inclined to use them. Through automated ordering, order tracking, and digital payment processing, the AWMC Ordering System promotes convenience. The AWMC Ordering System's implementation of safe and user-friendly features is strongly supported by the study done by Pavlou and Fygenson.

Loyalty Programs Research

Empirical studies on loyalty programs demonstrate that tiered reward structures increase customer retention by escalating benefits with repeated purchases. These incentives cultivate long-term engagement, as accumulating rewards motivate sustained patronage. The AWMC Ordering System's tiered membership model (Silver to Diamond) mirrors this approach, offering progressive rewards to distributors and customers for enduring loyalty. Also, tiered membership programs improve connections between companies and clients. Consumers are more inclined to stick with a business and refer others to its items if they feel appreciated. Customer involvement is another key idea included in studies of loyalty programs. Customers are encouraged to engage with the platform on a regular basis and take part in promotional events by reward schemes. By providing special offers, membership incentives, and personalized rewards for current members, the AWMC Ordering System encourages consumer participation. These studies emphasize how crucial membership programs are to increasing client retention and business expansion.

2.3 Local Literature

The Department of Information and Communications Technology (DICT) is the primary government agency mandated by Republic Act No. 10844 to formulate and implement policies, plans, and programs for information and communications technology (ICT) in the Philippines. Under this mandate, the DICT leads the Philippine Digital Transformation Strategy 2022, which aims to leverage ICT to improve government services, strengthen the economy, and enhance the quality of life of citizens, with a particular focus on enabling small and medium enterprises (SMEs) to adopt digital technologies.

The DICT's responsibility spans 82 provinces, approximately 1,700 local government units, around 115 million citizens, about 2 million SMEs, and more than 42,000 barangays, underscoring the scale and importance of promoting digital adoption nationwide, especially among medium-sized enterprises such as AWMC.

In the context of direct selling, data from Agriculture and Agri-Food Canada indicate that the value of direct selling in the Philippines reached approximately 1.37 billion US dollars in 2021, with projections suggesting growth to about 1.75 billion US dollars by 2026. These figures highlight both the current significance and the growth potential of the direct-selling sector in the country, reinforcing the need for robust digital systems that can support multi-level distributor structures and monitor performance effectively. Within this policy and market context, initiatives like the DICT's programs and the Philippine Digital Transformation Strategy 2022 are crucial in supporting companies such as AWMC in adopting and optimizing digital ordering and management systems.

2.4 Local Studies

Medium-sized enterprises in the Philippines account for approximately 99.5% of registered businesses, generate about 65% of total employment, and contribute around 25% of export revenues, according to the Joint SDG Fund (2024). Despite their economic significance, many of these businesses continue to experience productivity challenges and low levels of digital adoption. The use of tools such as cloud-based platforms and digital payment systems enables firms to automate processes like inventory management and order processing, thereby reducing labor requirements, minimizing errors, and lowering operational costs.

In the direct-selling industry, distributor retention remains a key challenge. Research cited by Triple i Consulting (2025), drawing on Euromonitor International, notes that distributors frequently leave due to inadequate training and insufficient income. To address this, companies are encouraged to implement well-structured incentive schemes and compensation models. Tiered membership programs are identified as an important mechanism for motivating distributors, encouraging sustained engagement, and fostering long-term loyalty.

These Philippine-based findings support the design choices embedded in the AWMC Ordering System. Its core features—including tiered membership levels, real-time inventory management, multiple digital payment options, and an analytics dashboard—align with the identified needs and priorities of local businesses.

Consequently, the AWMC Ordering System is well-suited to the Philippine direct-selling environment and can play a significant role in helping similar companies improve efficiency, retain distributors, and support sustainable growth.

2.5 Synthesis of the Reviewed Literature and Studies

The reviewed foreign and local literature consistently indicates that internet-based ordering and inventory management systems are highly beneficial for businesses that depend on memberships or distributor networks. These systems help organizations save time and reduce costs while improving customer satisfaction through faster, more accurate service.

Scholars such as Turban et al. and Laudon and Laudon emphasize that digital platforms minimize errors, increase operational efficiency, and strengthen data-driven decision-making. Pressman and Maxim highlight software engineering practices—such as modular design, iterative development, and scalable architectures—that enable systems to evolve and handle growth effectively. Pavlou and Fygenson further show that users are more likely to complete online purchases when they trust the payment process and perceive the system as secure, useful, and easy to use; this insight directly guided the design of the AWMC Ordering System's payment features.

Locally, government initiatives on digital transformation prioritize enabling small and medium enterprises to adopt ICT solutions, while industry data demonstrate the strong potential of membership-based and direct-selling businesses in the Philippines. Together, these sources define the functional and non-functional requirements that the AWMC Ordering System must address.

In response, the AWMC Ordering System was developed using PHP and MySQL and incorporates key features such as multi-level membership management, multiple payment options, real-time inventory tracking, and reporting tools to support managerial decision-making. It is specifically tailored to the context of a Philippine direct-selling company while adhering to internationally recognized best practices in e-commerce and software engineering.

Overall, the system operationalizes the concepts and recommendations from the reviewed literature by facilitating distributor engagement, streamlining the ordering process, and supporting the long-term growth of Amazing World Marketing Corporation.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study focuses on the design, development, testing, and evaluation of the Amazing World Marketing Corporation (AWMC) Ordering System. It employs a developmental or systems development research design, which is appropriate because the study involves creating a functional web-based application that addresses real operational problems in ordering and inventory management. The AWMC Ordering System is systematically evaluated to determine whether it performs reliably, is user-friendly, and meets the defined functional requirements.

3.2 System Development Methodology

The Amazing World Marketing Corporation (AWMC) Ordering System will be developed using the Agile Software Development methodology. This approach enables the five-member development team to work iteratively, refine features based on continuous testing and stakeholder feedback, and adapt to changing requirements throughout the project.

The development process will follow these phases:

- Requirements Analysis – Based on AWMC's operational requirements, the development team collects and records all system requirements. This entails creating technical specifications, defining system functionalities, and allocating user roles. To comprehend the organization's current procedures and difficulties, data collection techniques like observations, interviews, and document analysis may be employed. A comprehensive list of user stories, functional requirements, and non-functional requirements-which form the basis for system development-are the results of this phase.
- System Design – Making a blueprint for the suggested system is part of the system design phase. This entails creating the user interface, database structure, and overall system architecture. To see how the system will work and how users will interact with it, the team creates wireframes and prototypes.

To show how data flows through the system and how various components are connected, entity-relationship diagrams (ERDs) and data flow diagrams (DFDs) are also made. This stage guarantees that the system is scalable, well-organized, and compliant with the specified requirements.

- Development – The front-end and back-end components will be implemented in short sprints, with regular code reviews and incremental testing to ensure proper integration and functionality. During the development phase, the system is built using appropriate technologies, including HTML, CSS, and JavaScript for the front-end, and PHP and MySQL for the back-end. The development process is carried out in short iterations or sprints, allowing the team to focus on specific features such as user authentication, product ordering, inventory management, membership functions, and payment integration. Regular code reviews and incremental testing are conducted to ensure that each component functions correctly and integrates smoothly with other parts of the system. This iterative approach helps in identifying issues early and maintaining high-quality code standards.

- Testing – Unit testing, integration testing, and user testing will be conducted to verify that the system is functional, secure, and user-friendly. User acceptance testing involves actual users interacting with the system to evaluate its usability and functionality. Security testing is also performed to protect sensitive data, particularly in areas such as user authentication and payment processing. Performance testing ensures that the system can handle multiple users and transactions without significant delays.

- Deployment (Prototype) – A functional prototype of the AWMC Ordering System is implemented and shown to stakeholders for assessment during the last stage. This prototype lets users experience the system's features in a real-world setting while showcasing its fundamental features. This stage's feedback is used to pinpoint problem areas and direct future developments. Before the system is fully implemented, the deployment phase acts as a validation step to make sure it satisfies user expectations and business requirements.

3.3 System Architecture

The Amazing World Marketing Corporation (AWMC) Ordering System is structured into three main layers: the Presentation Layer, the Application Layer, and the Data Layer. This layered architecture ensures clear separation of concerns, maintainability, and scalability.

The Presentation Layer serves as the front end of the system and is implemented using HTML5, CSS3, and JavaScript. It provides a consistent and user-friendly interface for both customers and administrators. For customers, this layer allows browsing of products, managing the shopping cart, checking out, and viewing order history. For administrators, it offers interfaces for managing products, orders, and users, as well as accessing an analytics dashboard. Responsive CSS design techniques are applied to ensure that the interface remains professional and usable across various devices and screen sizes.

The Application Layer functions as the back end of the system and is developed using PHP. It is responsible for core business logic, including user authentication and authorization, order management, stock verification, queue number generation, membership validation, and payment processing. This layer also incorporates security mechanisms such as protection against forged requests, session timeouts, input validation, and rate limiting on login attempts. Access control measures are implemented to ensure that users can only interact with the modules and data that correspond to their roles and permissions.

The Data Layer acts as the gateway to the database and is implemented using MySQL. It handles the storage and retrieval of all persistent data, including user accounts, product information, orders, and transactions. The database is designed with appropriate normalization, relational constraints, and indexing to maintain data integrity, ensure correct relationships among entities, and optimize query performance. These design choices help guarantee the reliability, efficiency, and security of data operations throughout the entire AWMC Ordering System.

3.4 System Features and Modules

The Amazing World Marketing Corporation (AWMC) Ordering System is composed of several core modules that work together to deliver a complete ordering and management platform.

Authentication Module

This module handles user registration, login, and logout, as well as session management. It secures user credentials, maintains active sessions, and ensures that only authenticated users can access protected system functions.

Product Management Module

This module allows administrators to add, update, and delete products. It supports three product categories: loose products, member-only products, and membership packages, and ensures that product information remains accurate and up to date.

Shopping Cart Module

This module creates and manages a shopping cart for each user. It verifies product availability, updates item quantities, and allows users to add or remove products before checkout.

Order Processing Module

This module manages the entire order lifecycle, from order placement to completion. It assigns a unique order number, records payment choices, processes payments, and updates product stock levels accordingly.

Membership Management Module

This module administers the AWMC membership tiers: Silver, Gold, Ruby, Emerald, and Diamond. It controls product access based on membership level and tracks each member's status and eligibility.

Analytics and Sales Reporting Module

This module provides administrators with a dashboard that displays key performance indicators, including revenue, sales trends, order statuses, payment methods used, and top-selling products.

User Management Module

This module enables administrators to view, search, filter, and modify customer accounts. It includes tools for updating membership status and managing user information.

Together, the Authentication, Product Management, Shopping Cart, Order Processing, Membership Management, Analytics and Sales Reporting, and User Management Modules form an integrated system that supports the full functionality of the AWMC Ordering System.

3.5 Database Design

Ordering System is one of the most important components of the project because it serves as the foundation for data storage, organization, retrieval, and management. The system utilizes MySQL as its relational database management system due to its reliability, scalability, security, and compatibility with PHP-based web applications. The database is designed to support efficient transaction processing, accurate inventory management, secure customer information storage, and effective reporting functions.

The smooth and efficient performance of the AWMC Ordering System depends on a well-structured database. The database enables the system to organize and store data about users, goods, transactions, payments, inventories, and memberships. The system minimizes redundancy, preserves data integrity, and facilitates safe

communication between various platform components by using relational database design concepts. The relational database gateway of the AWMC Ordering System is made up of a number of linked tables. To support the platform's overall operations, each table is created with certain functions and linkages. Users, Products, Orders, Order_Items, and User_Payment_Accounts are the database's main tables.

- **Users** - Customers, members, and administrators who use the AWMC Ordering System have all of their account-related data stored in the Users table. The main repository for user identity and authentication information is this table. To identify each user record, a unique user identification number is provided to each registered account in the system. Important variables including full name, email address, hashed password, phone number, delivery address, membership status, and user role are all included in the Users table. For identification reasons, the customer's or administrator's entire name is stored in the full name field. For alerts, order updates, and account verification, the email address is the main login password and communication route.

To improve security and protect user credentials from unauthorized access, hashing techniques are used to encrypt passwords stored in the system. Sensitive password information is safeguarded even in the event that the database is hacked. The contact number field keeps client phone numbers, enabling administrators and delivery staff to speak with customers about transactions and deliveries. The place where client orders will be delivered is noted in the delivery address section.

Another important component of the Users table is membership status. The user's membership category—Silver, Gold, Platinum, or Diamond—is indicated in this field. Discounts, rewards, and extra rewards are applied during transactions using membership information. Each account's level of access within the system is determined by the user role field. While administrators are given further powers to manage inventories, keep an eye on sales figures, and change transaction records, customers are simply permitted to carry out tasks like browsing items, placing orders, and maintaining profiles. Because it supports the system's membership, customer administration, authorization, and authentication functions, the Users table is crucial. By maintaining organized & centralized user records, it also helps with data security and organization.

- **Products** – All of the AWMC Ordering System's product the catalog information is in the Products database. Customers may explore available goods on the site while administrators can manage product listings thanks to this table. To set each product entry apart from other items in the database, a unique product identification number

is assigned. Important data such product name, description, price, type, picture file location, and current stock amount are all found in the Products table.

The official name of the item being sold appears in the product name column. Product descriptions provide customers full information on the characteristics, components, applications, and advantages of the product. Customers can use these descriptions to make well-informed purchases. Each product's selling price is recorded in the product price section. Depending on the customer's membership status, the system may additionally apply membership-based pricing or special discounts.

Products are categorized using the product type field into groups including package offers, member-only products, and loose things. The categorization allows the application of specialized promotions by administrators and enhances product organization. Product images uploaded to the system can be found in the image file path field. By offering visual representations of the products that are available for purchase, product photos improve the customer experience.

The number of products that are available in inventory is tracked by the current stock quantity column. Whenever an order is finished, this field is immediately updated, guaranteeing precise inventory management and avoiding overselling. Product categories and effective inventory management are supported by the Products table. Additionally, by offering precise product details and current stock availability, it enhances consumer satisfaction.

•Orders – Complete order transaction records generated within the AWMC Ordering System are recorded in the Orders database. The primary database for all client transactions and purchases is this table. A unique order identification number and an automatically generated queue number are assigned to every order transaction. The queue number guarantees the efficient processing of pickups and deliveries and aids in the organization of client transactions. Customer identification number, order method, payment method, payment status, queue number, total amount, and order status are all included in the Orders table.

The customer identification number links each order to the corresponding user account within the Users table. This relationship allows administrators to view transaction histories and customer purchase records. The order method field identifies whether the transaction is for delivery or pickup. This information helps administrators and staff prepare appropriate fulfillment procedures. The payment method field records the selected payment option used by the customer, including GCash, PayMaya, PayPal, Cash on Pickup, or Cash on Delivery.

Whether the order has been paid for, is pending, or has been finished is shown in the payment status column. Administrators can maintain an eye on financial activities and confirm payment completion with the aid of this function. The total price of the transaction, including product pricing and any relevant fees, is kept in the total amount column. Pending, processing, shipping, finished, and canceled transactions are all tracked via the order status column. Because it centralizes transaction administration and facilitates order tracking, payment monitoring, and sales reporting, the Orders table is crucial.

- Order_Items** – The exact product line items assigned to each client order are kept in the Order_Items database. By listing the particular things bought in each transaction, this table creates the link between orders and products. Order identification number, product identification number, quantity ordered, and unit pricing at the moment of purchase are among the data included in every record in the Order_Items table.

While the product identification number links the item record to the related product in the Products table, the order identification number connects the item record to the matching order in the Orders table. The quantity ordered field keeps updated with how many units a client has bought of a specific product. To provide accurate historical transaction records, the product price at the time of purchase is stored in the unit price column.

The Order_Items table supports accurate order processing and transaction documentation. It also helps administrators analyze sales trends by identifying frequently purchased products and customer buying behavior. The solution maintains effective relational data management and enhances database structure by separating into order details into separate line items.

- User_Payment_Accounts** – Customer payment account details for online payment services like GCash and PayMaya are kept in the User_Payment_Accounts database. For quicker and easier checkout procedures, customers can save and reuse payment information in this table. Payment account identification number, user identification number, payment platform type, account name, and account number are all included in each payment account record.

The payment account and the matching customer profile in the Users database are connected via the user identification number. This relationship ensures that the right account owner is linked to the stored payment details. The payment platform type field identifies whether the account belongs to GCash, PayMaya, or other supported digital payment services. The information needed for transaction processing and payment verification is stored in the account name and account number columns.

To stop unauthorized access to stored payment information, the system employs security measures. Secure transaction procedures are used while handling sensitive data, which is encrypted. By preventing the need to continually input payment information during each transaction, the User_Payment_Accounts database increases consumer convenience. Additionally, it facilitates quicker checkout processes and raises consumer satisfaction levels overall.

3.6 Development Tools and Technologies

- The five-member development team of the AWMC Ordering System utilizes various software development tools and technologies to design, develop, test, and maintain the system effectively. These tools were selected based on their compatibility, reliability, accessibility, and suitability for web-based application development.

The chosen technologies support both frontend and backend development, database management, analytics generation, version control, and testing processes. By combining these technologies, the development team can ensure efficient collaboration, reliable system performance, and organized software development practices.

- PHP: The primary server-side programming language used to create the AWMC Ordering System's middleware features and backend components is PHP. PHP was chosen because to its strong community support, simplicity of interaction with web technologies, and compatibility with MySQL databases. PHP is in charge of processing user requests, controlling sessions, managing authorization and authentication, verifying user input, and communicating with the database.

The language makes it possible for developers to produce dynamic content and responsive, interactive web apps. Also, PHP allows for modular development, which lets developers divide the system into distinct parts like order processing, product management, user administration, and analytics. PHP is a good choice for creating web-based ordering systems like AWMC because of its scalability and versatility.

- MySQL: MySQL serves as the relational database management system of the AWMC Ordering System. It is responsible for storing, organizing, retrieving, and managing all system data. MySQL was selected because of its effectiveness, dependability, security features, and PHP compatibility. Relational data structures

that minimize duplicates and preserve consistency are supported by the database. Also, MySQL facilitates effective data retrieval and querying procedures.

Customer accounts, product inventories, transaction records, membership data, and payment details are all stored by the system using MySQL. MySQL gives you accurate data management and safe storage, which enhances the AWMC Ordering System's performance and stability.

•HTML5, CSS3, JavaScript: The main client-side technologies utilized in creating the AWMC Ordering System's user interface are HTML5, CSS3, and JavaScript. Web pages' content and structure are provided by HTML5. Forms, product displays, navigation menus, and other interface elements are made using it. The platform's visual design and styling are handled via CSS3. Through layouts, colors, typography, and responsive design strategies, it enhances the system's look.

JavaScript gives the system dynamic behavior and interactivity. It makes features like product filtering, interactive menus, real-time form validation, and asynchronous content changes possible. When combined, these technologies produce a responsive and easy-to-use interface that enhances accessibility and consumer experience.

•Chart.js: The AWMC Ordering System's analytics dashboard uses the JavaScript package Chart.js to create charts and data visualizations. Through graphical representations, the library enables administrators to see sales statistics, transaction summaries, inventory reports, and customer trends.

By simplifying complicated information into visual representations, charts and graphs enhance data interpretation. Bar charts, line graphs, pie charts, and radar charts are just a few of the graphic formats that graphic.js provides. The system's analytical skills are enhanced and managerial decision-making is supported by the integration of Chart.js.

•XAMPP: The development team uses XAMPP, a local development environment, to test and prototype the AWMC Ordering System. The platform makes setting up a local web server environment easier by combining MySQL, PHP, and Apache into a single software package.

Before deployment, developers utilize XAMPP to test system features, troubleshoot issues, and assess performance. Because XAMPP allows the team to simulate actual server processes within a local computer environment, it increases development efficiency.

- Visual Studio Code: Visual Studio Code serves as the primary code editor used by the development team. The editor provides features such as syntax highlighting, code suggestions, debugging tools, and extension support that improve programming productivity. Visual Studio Code supports multiple programming languages and integrates with Git and GitHub for version control. The development team uses Visual Studio Code to write, edit, organize, and debug source code efficiently.

- Git and GitHub: For collaborative software development and version management, Git and GitHub are utilized. Git allows developers to manage changes and restore earlier versions when needed by keeping track of changes made to source code files. Team members may save, exchange, and work together on project files using GitHub, an online repository platform. Code disputes are avoided, cooperation is enhanced, and structured software development procedures are supported by using Git and GitHub. Additionally, these technologies allow developers to save backup copies of source code and record project progress.

3.7 Data Gathering Techniques

The development of the AWMC Ordering System requires accurate and reliable information regarding the organization's existing operations, customer needs, and system requirements. To gather the necessary information, the researchers employ several data gathering techniques during the requirements analysis phase.

These methods aid in defining the functional and non-functional needs of the suggested system, identifying operational issues, and figuring out user expectations. The study's main methods for collecting data were questionnaires, document analysis, observation, and interviews.

- Interviews are conducted with AWMC management personnel, staff members, and selected stakeholders to gather detailed information regarding the organization's existing ordering and inventory management processes. Researchers can ask straightforward questions and get thorough answers about operational methods, difficulties, and company needs through the interview process.

The researchers use interviews to identify current problems such as unsuccessful order tracking, inventory anomalies, and manual transaction delays. Customers' and administrators' expectations for the suggested system are also ascertained through interviews. In order to explain the features, functions, and goals of the AWMC Ordering System, information gathered through interviews is crucial.

•The current manual ordering and inventory management procedures of AWMC are observed. Through direct observation, the researchers keep an eye on how client orders are handled, transactions are processed, and inventory is tracked. The technique helps in discovering operational difficulties, repeated processes, delays, and inefficiencies in the current system.

Also, observation offers insights on customer interactions and staff procedures. Researchers can uncover chances for automation and process improvement by using the information acquired via observation.

•Reviewing existing business documents and records related to AWMC business operations is known as document analysis. Order forms, product catalogs, price structures, membership packages, inventory records, and transaction reports are some examples of these papers. By analyzing these materials, the researchers gain a better understanding of the organization's current procedures and data management practices.

The information that needs to be in the suggested database and reporting system is also found through document analysis. The technique confirms that the system requirements are accurate and comprehensive.

•Current AWMC members and clients are given surveys to complete in order to get input on their expectations, preferences, and experiences. Customer satisfaction, preferred payment methods, desired system features, and user interface preferences are all gathered through survey surveys. Researchers are able to efficiently collect data from numerous respondents using the survey procedure.

In order to make sure that the system meets the requirements and expectations of its intended users, customer input is crucial. The design of user-friendly interfaces, payment methods, and membership features is influenced by the survey results. All things considered, the combination of surveys, observation, document analysis, and interviews yields the thorough data required to create an efficient and trustworthy AWMC Ordering System.

3.8 Testing Procedures

Before being deployed, the Amazing World Marketing Corporation (AWMC) Ordering System will go through a number of thorough testing methods to make sure it operates effectively, safely, and dependably. Because it finds faults, confirms system operation, and guarantees that the program meets both technical and user requirements, testing is a crucial stage of software development. Through

methodical testing, the developers may assess if the platform can offer administrators and consumers a solid and user-friendly experience, as well as whether the system components function appropriately under various circumstances.

Unit testing, Integration testing, Security testing, and User Acceptance Testing (UAT) represent the various stages of the AWMC Ordering System testing process. Every testing stage contributes to the platform's overall dependability and efficacy by concentrating on a certain facet of system quality assurance.

•Unit Testing

The first phase of system testing carried out throughout the development process is called unit testing. The main goal of unit testing is to make sure that each module of the AWMC Ordering System operates in accordance with its intended functionality without relying on other system components. In this phase, individual PHP functions, modules, and database queries are tested independently to confirm that each component operates correctly when given different inputs and conditions. To find logical mistakes, inaccurate outputs, and unexpected behaviors, developers look at certain parts of the source code.

To check if the system accurately authenticates legitimate users and rejects incorrect login credentials, for instance, the login module is checked. In a similar vein, product management features are evaluated to confirm that products may be successfully added, altered, or deleted from the database. During unit testing, database queries are also assessed to make sure that records are properly added, updated, retrieved, and removed. Unit testing greatly improves system reliability by enabling developers to isolate and fix errors prior to integrating modules into the entire application. This process assists developers in identifying syntax errors, data inconsistencies, and processing issues early in the development cycle. The developers lessen the likelihood of system-wide failures in later phases of development by making sure each component operates correctly in isolation.

•Integration Testing

After unit testing, integration testing will be carried out to confirm that the various AWMC Ordering System modules and components function as a cohesive unit. The objective of integration testing is to guarantee that data flows reliably and effectively across the system. This testing phase focuses on verifying the communication and interaction between the front-end interface, middleware components, and database layers. Developers review if data input via the user interface is appropriately handled by the middleware and appropriately saved in the database throughout this procedure.

For example, the system must correctly receive product information, compute the total amount, create a queue number, store transaction data, and update inventory records when a customer places an order. Integration testing confirms that these linked processes are carried out error-free and uninterrupted. To ensure that transaction data is sent safely and properly, the integration of payment gateways like GCash, PayMaya, and PayPal is also checked. The analytics dashboard is also checked to make sure that real-time sales and inventory data are accurately shown in reports and charts.

Because modern online applications are made up of multiple interconnected elements, integration testing is crucial. When components interact with one another during unit testing, issues may still arise even if individual modules operate as intended. This stage guarantees that every component of the system functions effectively and harmoniously as a single platform.

•Security Testing

Security testing is performed to discover vulnerabilities and weaknesses that might affect the AWMC Ordering System's availability, confidentiality, and integrity. Strong security measures must be put in place since the system manages sensitive client data, payment information, and transaction records. Evaluating the system's capacity to shield user data from malicious activity, cyberattacks, and illegal access is the main goal of security testing. Finding typical web application vulnerabilities like SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF) is the main goal of the testing procedure.

To figure out if attackers may change database queries with malicious input, SQL injection testing is done. Before executing database transactions, developers make sure that all user inputs are verified and cleaned. Testing for cross-site scripting (XSS) examines whether malicious scripts can be inserted into websites and run by other users' browsers. To avoid this problem, appropriate input validation and output encoding techniques are used.

Testing for cross-site request forgery (CSRF) evaluates whether unwanted operations may be carried out without the agreement of authenticated users. To lessen this risk, session validation techniques and security tokens are employed. To improve account security, the system also uses session management and password hashing. While secure session handling guards against unwanted access to user accounts, encrypted passwords shield user credentials from disclosure. Security testing ensures that the AWMC Ordering System complies with secure software development methods and shields administrators and users from possible online dangers.

- User Acceptance Testing (UAT)**

The final phase of system testing before deployment is called user acceptance testing, or UAT. To attempt to determine whether the system meets operational objectives and user expectations, AWMC administrators, staff, and selected group of customers test the deployed prototype. UAT's main objective is to evaluate the system's usability, practicality, and reliability in a real-world setting. In contrast with technical testing phases, UAT focuses on end users' actual experiences with the platform.

3.9 Evaluation Criteria

Using a standardized assessment system, the AWMC Ordering System will be systematically evaluated according to the following standards. The evaluation's goal is to ascertain if the system achieves the required standards for effectiveness and quality. The system's overall effectiveness in handling customer transactions and administrative duties will be the main focus of the evaluation procedure. Also, the evaluation will identify possible shortcomings and areas in need of development. To make sure the system operates in accordance with user expectations, every criterion will be thoroughly checked. To ensure the ordering system's performance and dependability, proper assessment is crucial.

- Functionality:** The ability of a system to perform out its intended functions accurately and fully is referred to as functionality. All required activities must be carried out by the system successfully. The ordering method should make it simple for customers to browse the products. Users should have no trouble adding items to their carts thanks to the system. Also, customers must be allowed to change amounts and delete items as needed. From product selection to payment confirmation, the ordering procedure need to function precisely. Totals, discounts, and other fees must be accurately calculated by the system. The database should accurately store order information. Accurate transaction records and receipts should be produced by the system. Effective client account management should be possible for the administrator. After each successful transaction, the product inventory has to be automatically updated. The system should be able to facilitate the creation of reports for inventory and sales tracking. The administrator panel should be complete and packed with functions. Users should receive precise results from search functions in a timely manner. Notification messages should display once transactions are completed successfully. Whenever a wrong activity is identified, users should be guided by error messages. Inaccurate transactions and duplicate orders should be avoided by the system. The system's modules should all be able to communicate with one another. For each step, the system must reliably produce correct results. To make sure every feature works properly, functional testing will be carried out.

•Usability: In order to reduce the learning curve and training requirements, the interface must be simple to use and accessible for administrators and customers alike. The system should be simple for users to navigate. Instructions, buttons, and menus should all be organized and well labeled. A user-friendly environment for efficient transactions should be provided via the design. Orders should be completed quickly and without confusion by customers. Convenient management responsibilities should also be carried out by administrators. Mobile phones, tablets, laptops, and desktop computers should all be supported by the system. On all screen sizes, the interface should continue to be aesthetically pleasing and responsive. When mistakes are made, users should be guided by appropriate feedback messages. First-time users should also have access to help tools and prompts. The entire design ought to enhance the user experience. The system should provide clear feedback after every action performed by the user. Customers should receive confirmation messages after placing orders. Error messages should explain problems in a simple and understandable manner. Font sizes and colors should improve readability and visibility. The layout should remain consistent across all pages of the system. The system should work properly on desktop computers and mobile devices. Administrators should easily manage records using the dashboard. Training requirements for users should be minimal. The system should improve customer satisfaction through convenience and accessibility. User testing will help determine whether the interface is effective and easy to use. Overall usability should encourage users to continue using the system regularly.

•Security: Protection of data and system resources from unauthorized access can be referred to as security. Customer data must always be protected via the AWMC Ordering System. Passwords and usernames should to be required in order to access accounts. Passwords should to be securely stored in the database. Only authorized administrators should be able to access the system. Sensitive client data needs to be kept private and secure. For account verification, the system has to provide authentication elements. Hackers are not supposed to be able to access the database thanks to security measures. To prevent malicious attacks, input validation should be used. Incorrect login attempts must be appropriately detected by the system. Inactive accounts should be immediately logged out thanks to session management. Additionally, backup data need to be shielded from unwanted access. Unauthorized modifications and data manipulation should be prevented by the system. Accurate and secure handling of database transactions is required. Important data should be transferred via encryption techniques. Accurate user activity and transaction logs should be kept by the system. Vulnerabilities in the system should be detected through security testing. To improve system security, upgrades should be implemented on a regular basis. Administrators must to keep an eye out for any questionable activity on the system. Maintaining user confidence and trust will be aided by appropriate security measures.

•Performance: Performance emphasizes the system's speed, effectiveness, and stability when it is in use. Pages in the AWMC Ordering System need should load quickly and efficiently. While exploring items, users shouldn't have to wait a lengthy time. The system must react to user requests and orders right away. Even during peak hours, transactions should go without any issues. Multiple users should be able to access the platform at once with the system. Database queries must to run quickly and effectively. Product searches ought to yield results quickly. During continuous operation, the system need to stay steady. To prevent crashes, server resources should be appropriately controlled. Even with high workloads, the system should continue to respond quickly. Optimization techniques should to improve data processing speed. The system's performance should not be influenced by large volumes of data. Accurate and timely reports should be produced. The system's maximum capacity will be ascertained through performance testing. After temporary interruptions, the system should recover quickly fast. To maintain continuous service availability, downtime should be kept to a minimum. Transactions should be smooth and uninterrupted for customers. The system should continue to function consistently across various browsers and devices. To maintain efficiency, proper maintenance should be carried out on a regular basis. A dependable and efficient ordering experience should be facilitated by overall performance.

•Reliability: The system's consistency and dependability in performing tasks correctly is commonly referred to as reliability. Results from the AWMC Ordering System have to remain reliable every time. Orders should always be processed correctly, with no information missing. The database's transaction records must continue to be complete and organized. Even after extended usage, the system should to keep working smoothly. All actions and transactions must maintain data integrity. Important data should not be permanently lost thanks to backup methods. The system should be restored using recovery processes following unforeseen failures. Despite small technical problems, the system should continue to operate as intended. The system's steady performance throughout time will be guaranteed via reliability testing. Consumers should have faith that their orders are handled correctly. Errors in data entry and transactions should be minimized by the system. The system's reports must continue to be reliable and accurate. Correct stock levels should always be reflected in inventory records. Incorrect calculations and duplicate entries should be prevented by the system. System functioning shouldn't be negatively affected by updates or maintenance. The process of ordering must to be accessible to consumers at all times. Transaction data should be reliably stored by the system without corruption. Maintaining customer happiness and corporate effectiveness requires dependable performance. System problems should be found and fixed quickly with the use of ongoing monitoring. The AWMC Ordering System's overall dependability guarantees that it can efficiently support regular corporate activities.

To put it all up, the assessment criteria will be the basis for evaluating the AWMC Ordering System's efficacy and quality. To confirm the system's functionality, usability, security, performance, and dependability, several testing techniques will be applied. Whether the system satisfies the necessary requirements for implementation will be ascertained through the evaluation procedure. The evaluation's findings will also point out areas that need to be strengthened or improved. Customers and administrators can receive better service from a well-evaluated system. Accurate and efficient system operation is ensured by proper assessment. Additionally, it supports operational stability and data security. To guarantee the long-term efficacy of the system, ongoing testing and observation are required. Through automation and improved transaction management, the AWMC Ordering System is anticipated to enhance corporate operations. In the end, the assessment procedure will guarantee that the system consistently provides dependable and excellent performance.

Data Flow Diagram (DFD — Level 0)

The Level 0 DFD illustrates the high-level data flows within the AWMC Ordering System:

Customer → AWMC Ordering System → MySQL Database

Customer Data Flows:

- One of the main external elements in the system is the customer. Customers use the online interface to access the AWMC Ordering System. At first, they submit names, usernames, passwords, and contact data for registration. After receiving the registration details, the system verifies the data entered. The system saves the client data in the MySQL database following validation. In order to access their accounts, customers must additionally provide their login information. In order to handle authentication requests, the system verifies the username and password that are kept in the database. The customer can access the system if their credentials are legitimate. The system rejects invalid login attempts. The website's product listings are available for customers to browse. Customers are displaying product information in real time by the system when it has been received from the database. The search feature allows customers to look for items. Search requests will be handled by the system, which then gets relevant product data. Products may be chosen by customers and added to their shopping carts. Every time a consumer changes the quantity or removes an item from their cart, the details are updated. The system uses the database entries to verify product availability prior to order confirmation. This procedure guarantees that orders for unavailable items cannot be placed.

- During the checkout process, customers provide their order details. Before completing the transaction, the system receives and confirms the order details. The entire payment amount is computed automatically by the ordering system. Every order transaction that is successful generates a queue number. Together with transaction information, the produced queue numbers are kept in the database. The database also stores order information, including amounts, client information, and chosen items. Customers receive order confirmation notifications from the system after that. Through the UI, customers may see their order statuses and queue numbers

- Every time a consumer accesses their account, the system constantly gets transaction data from the database. Clients have access to transaction history and prior orders. Additionally, updated product data is regularly gathered from the database. During browsing, customers are provided with accurate product listings and stock information. Smooth and effective transactions are ensured by the interaction between consumers and the ordering system. Customer data is processed correctly and securely. Customer data flows support the system's orderly transaction management.

Administrator Data Flows:

- Products, transactions, and system operations are under the control of administrators. The admin dashboard provides administrators with access to the AWMC Ordering System. To access administrative features, they provide their login credentials. Before allowing access, the system verifies administrator accounts. Administrators can control product and client transactions after logging in. The dashboard interface is used by administrators to submit product changes. Product names, descriptions, prices, and stock levels may all be updated. After processing these modifications, the system saves the updated data in the MySQL database. The consumer interface instantly displays updated product data. Customers may always access accurate product information thanks to this real-time updating mechanism.

- Administrators use the system to handle consumer orders as well. Order statuses like pending, processing, finished, and cancelled can be altered. These status updates are received by the system, which then enters them into the database. After then, customers may utilize their accounts to see updated order statuses. Products that are no longer available or discontinued can also be deleted from the database by administrators. The system automatically refreshes the product listings and handles deletion requests. Administrators may keep an eye on sales and transaction activity using the admin dashboard. Every time a report is requested, the system pulls transaction data from the database. Transaction summaries, inventory reports, and sales reports can all be produced by administrators. These reports aid in tracking product availability and company performance. To avoid unwanted changes, all administrative activities are handled securely. In order to help with administrator

tasks, the ordering system maintains regular communication with the MySQL database. Consistency between the database and the user interface is preserved with the use of real-time updates. Every management activity that administrators take is instantly mirrored throughout the platform. Accurate inventory management and effective order processing are guaranteed by the administrator data flow. Supporting dependable system performance and well-organized company processes is facilitated by effective administrative control.

Entity Relationship Diagram (ERD)

The database structure utilized in the AWMC Ordering System is shown in the Entity Relationship Diagram (ERD). It illustrates the main entities, their characteristics, and the connections among them. The ERD helps efficient data organizing in the system database. Also, it guarantees accurate and methodical information storage. Database design and development are based on the diagram. Developers can understand connections between data in the system by using the ERD. Additionally, it aids in preserving the integrity and uniformity of documents. Effective querying, data retrieval, and transaction processing are all supported by the ERD. An essential part of the ordering system is represented by each entity in the database. Relationships between entities provide efficient data administration and communication.

Entities and Key Attributes

- **User:** The registered user of the AWMC Ordering System is symbolized by the User object. Account information for administrators and customers are stored by this entity. The primary key (PK) is a unique `user_id` that is issued to each user. The user's whole name is stored in the `name` property. The user's email address is saved in the `email` records for correspondence and account validation. For security reasons, the `password` property keeps hashed or encrypted passwords. The user's phone number is stored in the `contact_number` property. The customer's residence or delivery address is contained in the `address` attribute. Whether a consumer is a member or a normal customer is determined by the `membership_status` property. The role ownership indicates if the account is owned by an administrator or a customer. The date and time the account was established are stored in the `created_at` property. Account management and authentication depend on the User entity. Also, it supports consumer identification and transaction tracking. The database securely keeps all user data. Relationships and foreign keys are used to link the User object to other entities.
- **Product:** In the AWMC Ordering System, every sellable item is represented by the Product entity. A unique `product_id`, which serves as the primary key (PK), is assigned to each product. The product name is stored in the `product_name` property. Details and information about the product are provided by the `description` element.

Each item's selling price is stored in the price property. Whether the item is loose, member-based, or package-based is controlled by the product_type property. The file location of product photos that are shown on the website is stored in the image_path property. The amount of goods that are available in inventory is recorded by the stock_quantity property. The time and date the product was added to the system are stored in the added_at property. Product monitoring and inventory management are supported by the Product entity. Customers can also use the online interface to browse the items that are provided. Administrators update product information on a regular basis. Maintaining accurate product records contributes to effective business operations. Order transactions are linked to the Product object via the OrderItem entity.

- Order: Any purchase transaction in the system is represented by the Order entity. A distinct order_id, which functions as the primary key (PK), is assigned to each order. A foreign key (FK) associated with the User object is the user_id property. The name of the client who placed the order is stored in the customer_name property. The customer's delivery address is stored in the address property. For communication reasons, client contact information is stored in the contact_number property. Whether the order is for delivery or pickup is indicated by the order_method field. The customer's chosen payment method is stored in the payment_method property. Whether a payment is pending, paid, or unsuccessful is indicated by the payment_status property. The order's current state is monitored via the order_status property. The created queue number for transaction organization is stored in the queue_number property. The transaction's total payment is recorded in the total_amount property. The date and time the order was placed are stored in the order_date property. In transaction management, the Order entity is crucial. It aids in monitoring consumer purchases and order status. Order data is safely kept in the database for tracking and reporting needs.

- OrderItem: Each order transaction's individual products are represented by the OrderItem object. The primary key (PK) is a unique item_id that is assigned to every entry. A foreign key (FK) associated with the Order object is the order_id property. A foreign key (FK) associated with the Product entity is the product_id property. The amount property keeps track of how many items the buyer has ordered. Each product's price is recorded in the unit_price property. Multiple goods can be included in a single order thanks to the OrderItem object. Additionally, it aids in keeping thorough transaction records. Each order item is connected to a specific item and transaction. The organization supports accurate inventory control and order computation. Sales reports and transaction summaries are produced with the use of OrderItem data. Order item management enhances system tracking and monitoring. The Order and Product entities are connected via this entity.

- **UserPaymentAccount:** Users' stored payment methods are represented by the UserPaymentAccount object. The primary key (PK) for each payment account is a distinct account_id. A foreign key (FK) associated with the User object is the user_id property. The customer's payment account type is indicated by the account_type field. The payment account's registered name is kept in the account_name property. The payment account number or identity is kept in the account_number property. Whether the payment account is the user's default method is determined by the is_default property. Multiple payment methods are allowed to be saved by clients using the UserPaymentAccount object. It makes transactions more convenient in the future. Information about payment accounts is safely kept in the database. The organization facilitates quicker and more effective checkout procedures. Additionally, it keeps users' payment management organized

Relationships

- Within the database system, relationships define how entities interact with one another. The relationship between the User and Order entities is one-to-many (1:N). This implies that a single user can utilize the system to place several orders. However, only one user owns each order. The OrderItem entity and the Order entity are related in a one-to-many (1:N) fashion. This implies that several goods or line items may be included in a single order. Every order item is a part of a single order transaction. The relationship of the Product and OrderItem entities is one-to-many (1:N). This means that one product could show up in several order items from several transactions. However, each order item belongs to a single product. Also, it is an a one-to-many (1:N) link between the User entity and the UserPaymentAccount object. This implies that a single user can store several payment accounts in the system. There is only one user associated with each payment account.

- Information inside the database is arranged correctly thanks to the relationships between entities. Data consistency and integrity across linked tables are preserved with the use of foreign keys. Duplication and erroneous records are avoided by the relational structure. Additionally, it increases the effectiveness of transaction processing and data retrieval. Scalability is supported by the ERD as the system develops and grows. Maintaining dependable system performance is aided by appropriate database linkages. The AWMC Ordering System's database structure is implemented using the Entity Relationship Diagram as a reference. All things considered, the ERD guarantees correct transaction handling, effective data administration, and well-organized business processes inside the system.

This relational structure ensures data integrity, efficient querying, and scalability for AWMC's operations.

Use Case Diagram

The AWMC Ordering System's Use Case Diagram shows how humans and the system interact. It identifies the main actors and the tasks they can complete on the platform. The diagram helps by providing a clear and structured explanation of the system's functional needs. Also, it shows how administrators and customers interact with the different ordering system features. System operations and user responsibilities are presented visually in the Use Case Diagram. It aids users and developers in understanding the system's capabilities. The Customer and the Administrator are the two main actors in the AWMC Ordering System. Within the platform, each actor has particular responsibilities and rights.

Customer Use Cases

- One of the primary users in the AWMC Ordering System is the customer. Customers utilize the web-based user interface to communicate with the system. Account registration is one of the customer's main use cases. Customers can register by providing personal details including names, addresses, phone numbers, and email addresses. Before entering the registration information into the database, the system verifies it. Customers can use their credentials to log in after registering successfully. Secure access to client accounts and customized functionality is made possible by the login procedure. Account information can only be accessed by authorized users thanks to authentication.
- The technology allows customers to browse the items that are accessible. Product names, descriptions, costs, and pictures are all included in product listings. Customers may use the search function to look for items. Users are shown matched products that the system includes from the database. Before making a purchase, customers may examine comprehensive product information. Adding items to the shopping cart is another significant use case. Before checking out, customers can choose more than one product and change the quantity. Every time a modification is made, the cart information is immediately updated by the system. If a customer decides they no longer want to buy a goods, they can take it out of their basket.
- Another major customer use case is the checkout process. After finishing their selections, customers go on to the checkout process. The system automatically determines the order total upon checkout. Before submitting their orders, customers confirm them and supply shipping information. The system creates transaction records and verifies the order details. Orders are given queue numbers so they may

be properly organized and monitored. Following a successful transaction, order confirmation notifications are shown.

- The checkout procedure is another important consumer use case. Customers proceed to the checkout procedure after completing their selections. At checkout, the order total is automatically calculated by the system. Customers verify their orders and provide shipping details before submitting them. The system checks the order data and generates transaction records. Queue numbers are assigned to orders so they can be properly organized and monitored. Order confirmation notifications appear after a successful transaction.
- Customers can select from a variety of payment options that the system supports. GCash, PayMaya, PayPal, cash on pickup, and cash on delivery are among the available payment methods. The chosen payment method is processed by the system, and payment information is entered into the database. Payment accounts can also be saved by customers for further transactions. Convenience for future transactions is enhanced with saved payment methods. When needed, customers can modify or delete their stored payment accounts.
- Viewing order history and queue positions is another use case. Through their accounts, customers may keep an eye on the status and development of their orders. The system shows updated data after retrieving transaction records. Additionally, customers have access to payment details and past purchases. Transparency and client convenience are enhanced by this feature.
- Also, customers can make changes to their personal profiles. Editing names, contact details, addresses, and passwords are all part of profile maintenance. If the system allows it, customers can also modify their account settings or profile images. Maintaining correct and up-to-date consumer details is ensured by updating profiles.
- Customers can also buy membership packages using the AWMC Ordering System. Membership packages may offer particular benefits, discounts, or promotions. Through the system, customers may choose from a variety of membership packages and finish the purchasing process. The database is used to store and update membership data. This use case enhances user loyalty and engagement on the platform.

Administrator Use Cases

- The AWMC Ordering System's second primary actor is the Administrator. Administrators are in charge of monitoring business operations and maintaining the system. Visiting the administrator dashboard is one of the main use cases for administrators. Before allowing access to administration features, the system confirms the administrator's credentials. Sensitive system functionalities can only be accessed by authorized administrators thanks to authentication.

- Products in the system can be managed by administrators. Adding new goods to the database is part of product management. During the product creation process, administrators supply product names, descriptions, prices, stock quantities, and photos. Product data is stored by the system and shown on the consumer interface. Also, administrators have the ability to edit current product data as needed. Price adjustments, stock changes, and description changes are examples of product upgrades. These modifications are instantly reflected throughout the platform via the system.

- Removing unavailable or discontinued products is another important administrator use case. The dashboard helps administrators to remove goods from the database. Products that have been removed are no longer visible in customer product listings. Accurate inventory records and product availability are maintained thanks to this procedure.

- Managing customer orders is another responsibility of administrators. The dashboard interface allows them to see orders that have been placed. Customer information, items chosen, payment options, and order statuses are all included in order information. Order statuses like pending, processing, finished, or cancelled can be changed by administrators. Customer accounts instantly reflect updated statuses. This functionality enables effective transaction management and monitoring.

- Another responsibilities of an administrator is to manage user accounts. Customer account information kept in the database is accessible to administrators. In response to the consumer purchases or subscription modifications, they could adjust membership statuses. When needed, administrators can also delete unauthorized or dormant user accounts. Maintaining orderly and secure system operations is aided by effective user management.

- Also, the AWMC Ordering System provides administrators access to sales information and analytics dashboards. Using generated reports, administrators may keep an eye on transaction activity and sales performance. Important company data, like total revenue, product performance, and consumer activity, are shown on the analytics dashboard. Sales monitoring features allows administrators to determine which goods are the best sellers. Reports assist administrators in making well-informed choices about marketing plans and inventories.

- Also, administrators may keep an eye on consumer interactions and system utilization. For security and monitoring purposes, the system logs significant activities. The ordering platform may be efficiently managed with the use of administrative tools. These usage scenarios guarantee the system's constant smooth and efficient operation

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